

$$m[2,4] = \max \left\{ m[1,4], m[1,4-3] + 2 \right\} \begin{matrix} \max \{4+0, 2 \\ 3+2, 2 \} \end{matrix}$$

$$= \max \{ 1, 0+2 \} \quad \max \{ 3+0, 2 \}$$

$$P = \{ 2, 3, 1, 4 \} \quad W = \{ 3, 4, 6, 5 \} \quad W = 8$$

I II III IV I II III IV

M

W →

i ↓

		0	1	2	3	4	5	6	7	8
P[i]	W[i]	0	0	0	0	0	0	0	0	0
2	3	1	0	0	2	2	2	2	2	2
3	4	2	0	0	2	3	3	3	5	5
4	5	3	0	0	2	3	4	4	5	6
1	6	4	0	0	2	3	4	4	5	6

$$8 - 5 = 3$$

$$m[i, w] = \max \left\{ m[i-1, w], m[i-1, w-W[i]] + P[i] \right\}$$

$$X = \{ 1, 0, 0, 1 \} \quad X = \left\{ \begin{matrix} \times & \checkmark & \times & \checkmark \\ 1 & 4 & 3 & 2 \\ 6 & 5 & 4 & 3 \\ \text{III} & \text{IV} & \text{II} & \text{I} \\ 0 & 1 & 0 & 1 \end{matrix} \right\}$$